
Current Topics in Digital Philology —

Assignment 3: Historical Linguistics

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1 Exercise 1

Assuming unit costs for insertions, deletions, and substitutions (meaning that they all have a cost of 1), the absolute edit distance between *crocodile* and *cocodrilo* is 3:

1. *crocodile* → *cocodile* (deletion of “r”)
2. *cocodile* → *cocodrile* (insertion of “r”)
3. *cocodrile* → *cocodrilo* (substitution of “e”)

2 Exercise 2

2.1 a

Assuming unit costs for insertions, deletions, and substitutions, the absolute edit distance between *abab* and *baba* is 2. This can be achieved by first deleting the first “a”, and then inserting an “a” at the end of the string:

1. “abab” → “bab” (total cost: 1)
2. “bab” → “baba” (total cost: 2)

2.2 b

Assuming unit costs for insertions and deletions, while assuming a cost of $\frac{1}{3}$ for substitutions, the absolute edit distance between *abab* and *baba* is $1\frac{1}{3}$. This score can be achieved by substituting every character in the string, which, under these costs, is cheaper than the insertion and deletion performed in (a):

1. “abab” → “bbab” (total cost: $\frac{1}{3}$)
2. “bbab” → “baab” (total cost: $\frac{2}{3}$)
3. “baab” → “babb” (total cost: 1)
4. “babb” → “baba” (total cost: $1\frac{1}{3}$)

It can thus be concluded that the optimal alignments for (a) and (b) are not the same.

3 Exercise 3

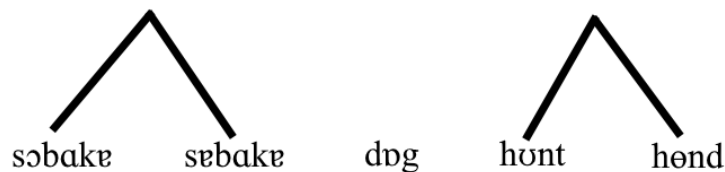
Skipped, as it is not part of the requirements for grade G.

4 Exercise 4

As a technicality, the missing word for "freshwater eel" in Awa Pit might cause issues with an automated cognate judgment program that assumes all cognates have values in all the provided languages. Assuming this is not an issue, I would expect that Cha'palaa and Guambiano words for "saltwater eel" would be marked as cognates, Cha'palaa and Awa Pit words for "shore" are marked as cognates, and that words in all three languages for "water" and "river" are marked as cognates, especially due to the partial cognate for "water" and "river" for Cha'palaa and Awa Pit specifically. Since we've already established these three languages to be part of the same Barbacoan language family, from a linguistic perspective, the ideal outcome would be for the automated cognate judgment to align with our predetermined linguistic classification, that is that words in all three languages would be considered cognates.

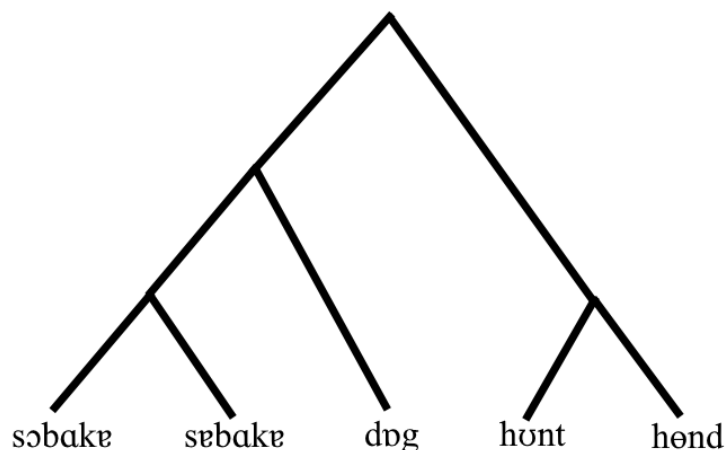
5 Exercise 5

5.1 a



The above figure shows a tree of words where a line between words indicates that they are cognates, assuming a threshold pairwise distance of $t = 0.25$, based on the original pairwise distance matrix. The more closely related two words are, the lower they split in the tree. $t = 0.25$ thus gives three cognate groupings (represented using non-IPA characters): {sobaka, sabaka}, {hunt, hond}, {dog}.

5.2 b



The above figure shows a tree of words where a line between words indicates that they are cognates, assuming a threshold pairwise distance of $t = 0.75$, based on the original pairwise distance matrix. The more closely related two words are, the lower they split in the tree. $t = 0.75$ thus gives one cognate grouping, which contains all the words of the original matrix. Although (non-IPA notation)

“dog-hunt” exceeds the threshold, “hunt-hond” does not, and since the property of cognacy is transitive, “dog-hunt” must also be cognates.

6 Exercise 6